

AMAR DHILLON

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Education

University of Maryland, College Park

B.S. in Computer Science, Minor in Statistics; GPA: 3.5

Expected May 2028

College Park, MD

Projects

IEX Order Book & Matching Engine | C++17, STL, POSIX, Linux perf

Jul 2026 – Aug 2026

- Built a C++17 order book and matching engine and replayed **50.9M** real IEX exchange messages through it without ever resyncing to the exchange's book, agreeing with the exchange on 99,122 of 99,123 fills and 100% of top-of-book prices
- Implemented B+ tree and LSM-tree backends for the event log and benchmarked them on identical 2M-record workloads, where the B+ tree gave **12.8x** the read throughput and 22x faster range queries at the cost of 27% more disk space
- Traced an apparent 1.65x slowdown to the CPU frequency governor using perf counters and a null control book, then cut run-to-run variance from about 6% to 0.06% with core pinning and a fixed clock
- Stored price levels in a std::map of std::list FIFOs with a hash map from order ID to list iterator for O(1) cancels, sustaining **1.26M events/sec** over the full replay and a 1.2 μs per-update p99 on a separate 2M-event run

Experience

Amazon Web Services

Sep 2026 – Dec 2026

Software Development Engineer Intern

Redmond, VA

- Writing Java logic that sets the upper and lower bounds on EC2 capacity that Amazon Redshift reserves for operations like patching, balancing availability against cost, and backtesting it

Capital One

Jun 2026 – Aug 2026

Software Engineering Intern

Richmond, VA

- Cut AI workflow generation time **65%** to under 15s by replacing a Claude skill's open-ended search with focused sub-skills that call Python tools for workflow search and validation, guided by CloudWatch and OpenSearch latency logs
- Because the 270K+ row PostgreSQL workflow table had strict access limits, precomputed popular workflows into a category tree that the Claude skill searches first, with Redis caching full workflows to cut database queries

UMD App Development Contracting (Client: Amazon Leo)

Jan 2026 – Present

Software Engineer (Contract)

College Park, MD

- Reduced end-to-end latency **34%** for a pipeline routing 6,000+ scraped research documents to inference by replacing S3 staging hops with Kinesis streams feeding Lambda

UMD Institute for Advanced Computer Studies (UMIACS)

Dec 2025 – Present

Machine Learning Research Intern

College Park, MD

- For a PhD-led robotics project aimed at ICLR, built a pipeline that overlays AnyGrasp's candidate grasps on camera images in Open3D for a vision-language model, then tuned prompts so the model picks the right grasp more often

TerpLabs

Jun 2025 – Present

Co-Founder, Software Engineer

College Park, MD

- Co-founded a 110+ member student product studio that has shipped 5 live products
- Built Redis caching in front of the Prisma ORM queries for Tortuga, the studio's class scheduler with 20,000+ visits in its first 30 days, to reduce database load and latency

Technical Skills

Languages: C++17, C, Python, Java, Rust, SQL, TypeScript, Bash

Systems & Performance: low-latency C++, debugging and profiling with Linux perf (hardware counters, top-down analysis), POSIX, STL, B+ trees, LSM trees, data structures and algorithms, benchmarking

Cloud & Tools: AWS (Lambda, Kinesis, EC2, Fargate, CloudWatch monitoring), AWS CDK, PostgreSQL, Redis, Docker, GitHub Actions (CI/CD), Git, Pytest

Certification: AWS Certified Developer – Associate (Nov 2024 – Nov 2027)